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0/1 Knapsack Problem Using Dynamic Programming

In [2]:

```python
n = int(input("Enter the number of items: "))

print("Enter the weight and profit for each item:")
weight = []
profit = []
for i in range(n):
    w = int(input("Weight of item {}: ".format(i+1)))
    p = int(input("Profit of item {}: ".format(i+1)))
    weight.append(w)
    profit.append(p)

capacity = int(input("Enter the capacity of the Knapsack: "))

def DP_Knapsack(capacity, weight, profit, n):
    K = [[0 for x in range(capacity + 1)] for x in range(n + 1)]

    for i in range(n + 1):
        for w in range(capacity + 1):
            if i == 0 or w == 0:
                K[i][w] = 0
            elif weight[i-1] <= w:
                K[i][w] = max(profit[i-1] + K[i-1][w-weight[i-1]], K[i-1]
            else:
                K[i][w] = K[i-1][w]

    return K[n][capacity]
max_profit = DP_Knapsack(capacity, weight, profit, n)
print("==================================================================
print("Maximum Profit Earned = ", max_profit)
print("==================================================================
```

```
Enter the number of items: 4
Enter the weight and profit for each item:
Weight of item 1: 2
Profit of item 1: 1
Weight of item 2: 3
Profit of item 2: 2
Weight of item 3: 4
Profit of item 3: 5
Weight of item 4: 5
Profit of item 4: 6
Enter the capacity of the Knapsack: 8
=====================================================================
Maximum Profit Earned =  8
=====================================================================
```